Building and Maintaining a Temporal Knowledge Graph for News Streams on China–Romania Relations

Andreea-Maria Onaci
Department of Computer Science
Technical University of Cluj-Napoca
Cluj-Napoca, Romania
andreea.onaci@campus.utcluj.ro

Mihaela Macarie-Ancău
Department of Computer Science
Technical University of Cluj-Napoca
Cluj-Napoca, Romania
mihaela.macarie@campus.utcluj.ro

Petruț-Betuel Paul
Department of Computer Science
Technical University of Cluj-Napoca
Cluj-Napoca, Romania
petrut.paul@campus.utcluj.ro

***Abstract*—We present a system for building and maintaining a temporal knowledge graph (TKG) from a multilingual news stream restricted to articles that mention both China and Romania. The pipeline ingests full article text from GDELT and curated RSS feeds, performs entity and relation extraction via spaCy and DeepKE, anchors temporal expressions to ISO 8601 intervals using HeidelTime TIMEX3 normalization, and stores the resulting graph in Neo4j. A bilateral-relations ontology grounded in CAMEO/GDELT event codes, IPTC Media Topics, and W3C OWL-Time governs the semantic structure of events and their validity intervals, with temporal relations between events computed under the full 13-relation Allen Interval Algebra rather than a coarse before/after/overlaps view. A credibility layer combines source-domain scoring with publisher-network-aware cross-source corroboration, embedding-based semantic claim merging, and NLI-based contradiction detection; a temporal gate grounded in the Allen relations, together with an explicit per-event-type claim-lifecycle state machine, distinguishes genuine factual contradiction from ordinary claim evolution (e.g. *proposed* → *signed* → *ratified*) before a claim is flagged as contested. Five competency questions drive the evaluation: tracking economic evolution, surfacing institutional linkages, enumerating bilateral events, identifying low-credibility narratives, and discovering collaborative research. We additionally report a rule-based reproduction of a coarse time-bucket temporal representation from prior fact-checking work as an ablation baseline for the Allen-relation gate. Modular architecture and reproducible schema migrations allow incremental update as new articles arrive, making the system suitable for ongoing monitoring of a bilateral news domain.**

***Index Terms*—temporal knowledge graph, news stream processing, bilateral relations, China–Romania, DeepKE, HeidelTime, TIMEX3, Neo4j, fake-news detection, GDELT, ontology reasoning, Allen interval algebra, natural language inference, claim-lifecycle verification**

I. Introduction

The diplomatic, economic, and cultural relationship between China and Romania has intensified significantly since Romania's accession to the European Union and China's Belt and Road Initiative expanded westward into Central and Eastern Europe. However, this bilateral domain is served by a heterogeneous, multilingual news ecosystem whose signals range from state-aligned outlets (CGTN, Xinhua) to independent Romanian business press and international wire services. Tracking the evolution of this relationship requires more than keyword monitoring; it demands structured, time-aware representation of events and the provenance of every claim.

Knowledge graphs (KGs) have become a standard representation for structured world knowledge, and temporal knowledge graphs (TKGs) [1] extend KGs with valid-time intervals so that facts can be queried relative to a point in time. Despite an extensive body of work on general-purpose TKGs (Wikidata, YAGO, EventKG [2]), there is no existing resource dedicated to the China–Romania bilateral domain, and general-purpose TKGs stop at storing a validity interval per fact: they do not reason about *how* two intervals relate beyond simple overlap, and they do not distinguish a claim that genuinely contradicts an earlier one from a claim that merely reports the next stage of the same real-world process.

This paper makes the following contributions:

- A modular, open-source pipeline that ingests full article text (not just headlines), deduplicates, and filters multilingual news articles, retaining only those co-mentioning China and Romania.
- A bilateral-relations ontology combining CAMEO/GDELT event codes, IPTC Media Topics, and W3C OWL-Time for temporal validity, with pairwise event relations computed under the complete 13-relation Allen Interval Algebra [15].
- Integration of HeidelTime TIMEX3 normalization and DeepKE for joint temporal and relational extraction.
- A credibility module that combines publisher-network-aware cross-source corroboration, sentence-embedding semantic claim merging, and NLI-based contradiction detection, gated by the Allen relations and an explicit claim-lifecycle state machine so that sequential reporting on an evolving story (e.g. a trade deal moving from *proposed* to *signed*) is not mistaken for a factual contradiction.
- A rule-based, same-mechanism-class reproduction of the coarse time-bucket temporal representation used in prior evidence-based fact-checking work [16], used as an ablation baseline to justify the added complexity of ex-

act interval-algebra reasoning over a bucketed-distance heuristic.
- Five competency questions evaluated against the resulting graph.

The remainder of this paper is organized as follows. Section II reviews related work, followed by an overview of the high-level system architecture in Section III. Sections IV and V detail the data ingestion and information extraction pipelines, respectively. Section VI presents the bilateral-relations ontology and its Allen-algebra temporal reasoning layer, which informs the graph schema and population discussed in Section VII. Section VIII introduces the credibility assessment layer, including semantic corroboration, NLI contradiction detection, and claim-lifecycle verification. Section IX evaluates the system using targeted competency questions and component-level ablations, and Section XII concludes the work.

II. Related Work

A. Temporal Knowledge Graphs

Temporal KGs extend the standard subject–predicate–object triple with a validity interval, enabling time-scoped queries. Representative works include T-REx [9], TimePlex [10], and KnowEvolution [1]. EventKG [2] is the closest general-purpose reference for our work, aggregating events from Wikidata, DBpedia, and YAGO with linked temporal metadata. None of these resources targets a bilateral news-stream context, incorporates a live credibility assessment layer, or reasons about interval relations beyond storing a validity range per fact. Qualitative reasoning about how two intervals relate to each other is a much older, separate line of work, formalized by Allen's Interval Algebra [15]: 13 mutually exclusive, jointly exhaustive relations (*before*, *meets*, *overlaps*, *during*, *starts*, *finishes*, *equals*, and their inverses) between two closed intervals. We adopt this formalism directly for event-pair reasoning in our graph (Section VI) rather than an ad hoc coarse bucketing of relations, and use it as the temporal gate that separates genuine contradiction from claim evolution in the credibility layer (Section VIII).

B. News-Driven Knowledge Graph Population

The Global Database of Events, Language and Tone (GDELT) [3] extracts structured event records from global news at large scale using the CAMEO event taxonomy [4]. GDELT provides rich provenance metadata but lacks fine-grained temporal normalization and bilateral-domain ontological grounding; its Full-Text Search API in particular returns only a title and, in practice, rarely a usable summary snippet, so a naive integration risks operating on headline text alone (Section IV).

NewsKG and similar systems [14] populate KGs from news corpora but do not maintain temporal intervals or include credibility scoring. Our system builds upon GDELT as an article-discovery backend while adding full-text extraction, ontology-aligned extraction, and provenance-aware credibility reasoning.

C. Information Extraction Tools

DeepKE [5] provides unified deep-learning models for named-entity recognition (NER), relation extraction, and event extraction. HeidelTime [6] extracts and normalizes temporal expressions to the TIMEX3 standard [7], which is critical for grounding event validity. We combine both tools with a rule-augmented spaCy pipeline to maximize recall on China–Romania domain-specific entities, and with trafilatura [19] for boilerplate-free full-text extraction from raw article HTML.

D. Fake-News, Contradiction, and Credibility Assessment

Automated credibility assessment of news claims is an active area of research [11]. Approaches range from source-domain reputation scoring to graph-consistency checks [12] and cross-document corroboration. A related but distinct line of work studies *implicit temporal reasoning* for fact-checking: Allein *et al.* [16] learn a time-bucket embedding per distance-in-days quantile between an evidence document and a claim and feed it as a feature to a neural verification model, rather than computing an exact qualitative relation between the two. Our credibility module integrates source-domain scoring, publisher-network-aware corroboration, and cross-document contradiction detection in a lightweight layer that does not require external fact-checking APIs at inference time, and Section IX reports a direct ablation of exact Allen-relation gating against a rule-based, same-mechanism-class reproduction of Allein *et al.*'s coarse time-bucket representation.

Contradiction detection between claim sentences is framed here as a natural language inference (NLI) problem, using a pretrained cross-encoder built on DeBERTaV3 [18]; semantic corroboration matching uses mean-pooled sentence embeddings from a MiniLM-based sentence encoder [17].

III. System Architecture

Figure 1 provides a high-level view of the pipeline. The system is organized into six modules:

- Ingestion
- Extraction
- Temporal Anchoring
- Ontology Annotation
- Graph Population
- Credibility Assessment

Intermediate results are persisted in SQLite (for articles and NLP outputs) before being loaded into Neo4j (for graph queries). All configuration resides in a single YAML file overlaid by environment variables, enabling secret-free reproducibility.

The pipeline is designed for incremental operation: each module can be rerun independently, new articles are deduplicated before processing, and numbered SQL migrations make the SQLite schema reproducible across deployments.

IV. Ingestion and Relevance Filtering

A. Article Sources

The system monitors two complementary article streams:

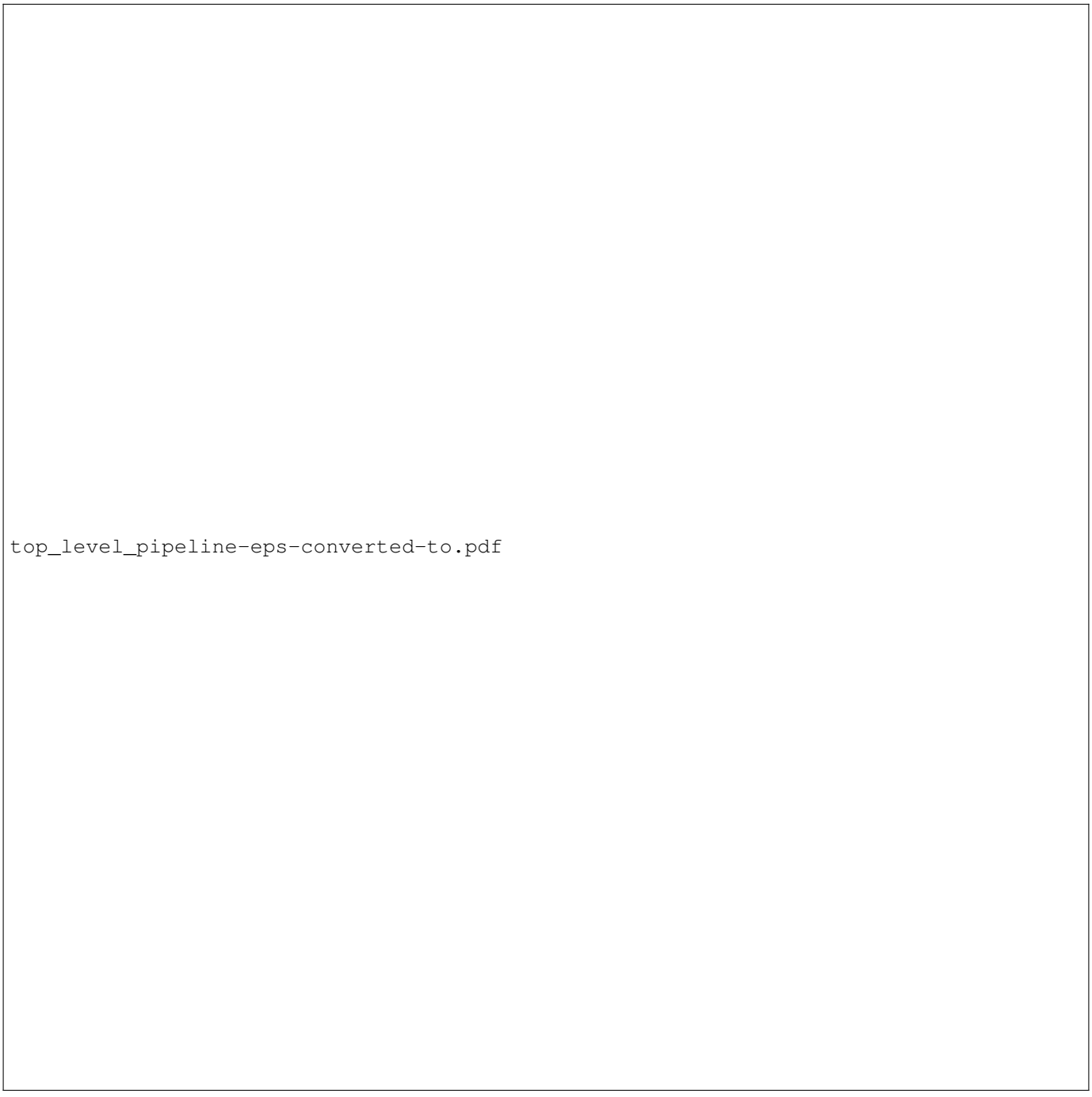


Fig. 1. Top-level orchestration and data flow of the Temporal KG pipeline.

a) GDELT Full-Text Search API: The primary source is the GDELT 2.0 Full-Text Search API [3] queried with the term pair `"China" AND "Romania"`. GDELT returns up to 250 article stubs per request, each containing a URL, title, and publication timestamp; in practice the API almost never populates a usable summary snippet for these stubs. GDELT provides the default credibility score of 0.7 assigned to each article prior to domain-level adjustment.

b) Curated RSS Feeds: Supplementary feeds are polled at configurable intervals (Table I). Feed-level metadata includes a source-country tag and a prior credibility score used by the credibility module.

TABLE I
RSS FEEDS MONITORED BY THE INGESTION MODULE

Feed	Focus	Credibility Prior
Google News (bilateral)	General	0.60
Radio Romania International	Political/cultural	0.80
Bursa.ro (capital markets)	Financial	0.60
Radio Free Europe	Political	0.80

B. Pipeline Steps

Each article stub passes through a nine-step pipeline before persisting to SQLite. A revision from an earlier version of this pipeline is worth stating explicitly, since it materially changes what every downstream module operates on: article text is now *fetched*, not assembled from title and snippet alone.

1) **URL deduplication**: SHA-256 hash of the URL is checked against the seen-URLs table; duplicates are skipped immediately.
2) **Full-text scraping**: the article URL is downloaded and its main body text is extracted with trafilatura [19] (boilerplate removal: navigation, related-links, and comment sections are stripped). This step is best-effort: dead links, paywalls, and transient failures fall back to title text rather than dropping the article, and every failure mode is logged rather than silently degrading. GDELT's Full-Text Search API returns only a title and (in practice) no usable snippet, so without this step the "article text" available to every downstream module – NER, relation extraction, temporal extraction, and corroboration – is just the headline. We verified this was in fact happening on the full ingested corpus prior to this fix (Section XI).
3) **Text assembly**: title and scraped body text (or, on scrape failure, title and any GDELT-provided snippet) are concatenated into `content_clean`.
4) **Language detection**: Detected using the `langdetect` library; language tag stored with the article.
5) **Translation**: Non-English articles are translated to English via a Google-Translate-backed client (`deep-translator`), chunked to the endpoint's per-request character limit, with a fixed inter-request delay and a defensive check against the endpoint returning an HTML error page as if it were a translation (Section XI), preserving the original text in `content_raw`.
6) **Content hashing**: SHA-256 of the normalized text enables a second, content-level deduplication pass.
7) **Relevance filtering**: The article must contain at least one token from the China alias set and at least one from the Romania alias set; irrelevant articles are discarded. A narrow secondary exclusion additionally drops otherwise-relevant articles that trip a sports-competition marker (competition-governing-body acronyms and explicit sports-section tags). This was added after full-text scraping surfaced a real failure mode: a genuine China-vs-Romania table-tennis match report satisfies plain co-occurrence, but sports results are out of scope for this project's 11-type bilateral *relations* ontology, and a multi-match roundup article can leak an unrelated match's sentence into event extraction. On the current corpus, this excluded 8 of 112 previously-relevant articles (three table-tennis reports, a judo report, and coverage of the World Aquatics Championships), and eliminated one of the corpus's 17 contradictory claim groups that traced directly to this failure mode.
8) **Persistence**: Relevant articles are written to SQLite and optionally exported as JSON.
9) **Deduplication registration**: URL and content hashes are registered in memory for future deduplication within the same run.

This two-stage deduplication (URL hash first, content hash second) reduces redundancy from syndicated content while preserving the same story covered by independent outlets, which is important for corroboration. It is, however, not sufficient on its own to make cross-outlet corroboration counts meaningful, since many outlets mirror the exact same wire-service copy rather than independently reporting it; Section VIII describes how the credibility layer corrects for this.

C. Large-Scale Corpus Build: GDELT GKG via BigQuery (2023)

To test the pipeline beyond the Full-Text Search API's 250-stub window, we selected candidate articles for calendar year 2023 directly from the GDELT GKG 2.0 table in BigQuery. The query keeps an article only if both `#RO#` and `#CH#` appear at least twice among its FIPS location codes and it carries no sports theme. This produced 19,996 unique URLs, which we ran through the full scrape, translate, filter, and store pipeline, using a local NLLB-200 (distilled, 600M) model for translation. The build took about 11 hours over five resumable runs, roughly 2 s per URL end to end (Table II).

a) Relevance precision at scale: Co-occurrence relevance is far less precise on this corpus than on the targeted searches used elsewhere in this paper. Only 693 of the 2,010 relevant articles (34%) name both countries in the same sentence. In a manual audit of 40 randomly sampled relevant articles, only 3 (about 8%) are actually about China–Romania relations (Table IV). Most are lists of countries, or stories about one country that mention the other in passing. The audit also revealed three concrete failure modes:

TABLE II
GDELT GKG 2023 CORPUS BUILD: OUTCOME

Stage	Articles	Share
Candidate URLs (BigQuery)	19,996	–
Stored after deduplication	18,002	100%
full text scraped	8,438	46.9%
scrape failed (title only)	9,564	53.1%
Relevance-filter decision		
no China/Romania term	11,391	63.3%
China only	3,845	21.4%
Romania only	497	2.8%
sports exclusion	205	1.1%
translation failed	54	0.3%
relevant (co-occurrence)	**2,010**	**11.2%**

TABLE III
SCRAPING FAILURES IN THE 2023 CORPUS BUILD

Failure	Articles
HTTP 403 Forbidden (anti-bot)	5,104
Connect timeout	916
Connection error	703
No extractable text	701
HTTP 404 Not Found	629
Read timeout	77
HTTP 429 Too Many Requests	54
Translation failed	54

1) the case-insensitive alias `ro` matches unrelated tokens (e.g., the Korean name "Ro"), and is the *only* Romania match in 180 relevant articles (9%);
2) 238 relevant articles still contain chess, Olympic, championship, or tournament vocabulary that the sports-marker exclusion does not cover;
3) the 600M NLLB model often produces fluent but unrelated output on Chinese pages (e.g., a headline about a livestreaming platform losing followers becomes "The South Korean government has announced that it will stop its operations"), while Russian output in the same sample was faithful. Among scraped articles longer than 500 characters, 18% of *no-match* articles show degenerate repetition (more than 30% repeated 4-grams), against 1% of relevant ones.

Most of the 63% *no-match* share is explained by scraping failure (8,902 of 11,391 had only a title). The remaining 2,489 scraped pages are largely domestic Chinese news, which shows that GDELT's automatic `#RO#` geotagging produces false positives even with the two-mention threshold.

b) What the corpus covers: The relevant set covers all of 2023, with 106 to 223 articles per month (Table V). The Romanian press is the largest national block (334 `.ro` domains). The context is dominated by the war in Ukraine: 1,017 articles mention Russia and 801 mention Ukraine, compared with 56 for the Belt and Road / 16+1 framework and 39 for Huawei/5G. In 2023, the two countries co-occur mainly as actors around the war (Romania as a NATO frontline state, China as Russia's partner) rather than through direct bilateral events. These findings motivate a sentence-level co-mention requirement, or a learned relevance classifier, before entity and event extraction is run on this corpus.

TABLE IV
MANUAL AUDIT OF 40 RANDOMLY SAMPLED "RELEVANT" ARTICLES

Category	n
About China–Romania relations	3
Country lists (rankings, reports, casualty lists)	13
One-country story, incidental mention	14
Calendars, almanacs, currency tables	4
Sports/chess not caught by exclusion	3
US politics (Biden-family business)	2
Alias false positive (`ro`)	1

TABLE V
RELEVANT ARTICLES PER MONTH, 2023

	J	F	M	A	M	J	J	A	S	O	N	D
Co-occur.	171	165	110	129	219	151	182	106	201	148	205	223
Same sent.	53	46	30	30	91	62	67	37	78	59	64	76

V. ENTITY, RELATION, AND TEMPORAL EXTRACTION

A. Named Entity Recognition

Entity extraction follows a *rule-first, model-augmented* strategy. The rule layer handles domain-specific surface forms with high precision, while the model layer provides recall on unseen names.

a) Rule-based extraction: Four deterministic passes operate on each sentence:

- **Country mentions**: alias tables for "China" (PRC, Chinese, People's Republic of China) and "Romania" (*România*, Romanian) with regex boundary matching.
- **Location mentions**: a curated gazetteer covering major Chinese and Romanian cities (Beijing, Bucharest, Shanghai, Cluj-Napoca, Constanța, etc.).
- **Person mentions**: capitalized multi-word tokens following a title keyword (President, Prime Minister, Ambassador, Ministrul, Ambasadorul).
- **Organization mentions**: capitalized spans containing organizational keywords (Ministry, Embassy, University, Bank, Council) or abbreviation patterns.

Each mention is assigned a confidence score: 1.0 for country tokens, 0.93 for title-anchored persons, 0.90 for keyword-matched organizations, and 0.74 for heuristic capitalized-span persons.

b) DeepKE integration: When DeepKE [5] is enabled, the system invokes a configured command template, writes a per-article JSON input, and reads the structured output. DeepKE mentions are merged with rule-based mentions using a span-deduplication key `(start_char, end_char, entity_type)`.

c) spaCy model augmentation: If a spaCy language model is available (`en_core_web_sm` by default), it provides sentence boundaries and can be extended with domain-adapted NER components.

B. Relation Extraction

Relation extraction maps entity pairs within a sentence to predicate types drawn from the bilateral ontology (Section VI). Pattern-based rules cover the most frequent surface realizations (e.g., *"X signed an agreement with Y"*, *"X invested in Y"*); DeepKE's relation-extraction module supplements these with learned patterns.

C. Temporal Expression Extraction

Temporal extraction relies on HeidelTime [6] via the `py-heideltime` Python package. HeidelTime annotates text with TIMEX3 tags and normalizes them to ISO 8601. Table VI shows the TIMEX3 types handled.

TABLE VI
TIMEX3 EXPRESSION TYPES HANDLED BY HEIDELTIME

Type	Example	ISO Normalized
DATE	"12 May 2026"	`2026-05-12`
DATE	"last week"	`2026-W22`
DURATION	"for three months"	`P3M`
SET	"every year"	`XXXX`
TIME	"at 14:00"	`T14:00`
INTERVAL	"valid until 2030"	`2026-05/2030`

When `py-heideltime` is unavailable (e.g., Java or Perl not installed), the system falls back to a hand-crafted regex engine covering the most common absolute date patterns. Extracted temporal expressions are linked to event nodes in the graph as `validFrom`/`validTo` interval endpoints, following W3C OWL-Time semantics, and feed directly into the Allen-relation reasoning described in Section VI-C.

VI. BILATERAL-RELATIONS ONTOLOGY

A. Event Taxonomy

The ontology defines 11 bilateral event types grounded in three established classifications (Table VII).

B. Temporal Validity Model

Each event node carries:

- `validFrom` – earliest ISO 8601 date extracted from the reporting article or TIMEX3 tag.
- `validTo` – latest bound, or `null` for open-ended intervals.
- `temporalGranularity` – one of {day, month, year}.

This models the W3C OWL-Time Interval and Instant classes [8] and aligns with the EventKG temporal representation [2].

combined_extraction_sequence-eps-converted-to.pdf

Fig. 2. Sequence diagram of the combined extraction module.

C. Allen-Relation Reasoning Between Events

Storing a validity interval per event is necessary but not sufficient for answering questions like *"did these two reports describe the same occurrence, or two different stages of an evolving story?"* We compute the exact qualitative relation between every pair of event intervals under Allen's Interval Algebra [15]: 13 mutually exclusive, jointly exhaustive relations (*before*, *after*, *meets*, *met_by*, *overlaps*, *overlapped_by*, *starts*, *started_by*, *during*, *contains*, *finishes*, *finished_by*, *equals*) between two closed intervals $A = [s_1, e_1]$ and $B = [s_2, e_2]$. A point-in-time event (a single reported date, the common case in news reporting) is represented as a degenerate interval with $s = e$.

An earlier version of this module collapsed the algebra into five coarse buckets (*before*/*after*/*contains*/*during*/*overlaps*), silently merging *meets* (two intervals that share exactly one boundary point) into *before*, and *equals* into *overlaps*. This under-serves a temporal-KG reasoning claim: two same-day reports of an event and two reports of genuinely overlapping-but-distinct events are a materially different relationship, and conflating them discards information a downstream consumer – including the credibility layer's temporal gate, Section VIII – would need. The full 13-relation algebra is now computed directly (`src/temporal/reasoner.py`); the original 5-bucket view remains available via an explicit coarsening function for call sites that only need it. A single ambiguity in the algebra's boundary cases is resolved deterministically: a point event whose date coincides with another interval's start simultaneously satisfies both *meets* and *starts*; our implementation checks full equality first (two point events on the same date are *equals*, not *meets*), then touching-boundary relations,

TABLE VII
BILATERAL EVENT TAXONOMY WITH CAMEO, GDELT, AND IPTC MAPPINGS

Event Type	CAMEO Code	GDELT Category	IPTC Topic
DiplomaticMeeting	04 Consult	Diplomatic Cooperation	Politics/Diplomacy
TradeAgreement	07 Provide Aid	Economic Cooperation	Economy/Trade
InvestmentProject	08 Yield	Economic Investment	Economy/Investment
TechnologyCooperation	05 Diplomatic Cooperation	Science and Technology	Science/Research
InfrastructureProject	07 Provide Aid	Infrastructure	Economy/Construction
EducationCooperation	05 Diplomatic Cooperation	Education	Education
SecurityStatement	13 Threaten	Security	Crime/Defence
SanctionOrRestriction	11 Sanctions	Sanctions	Politics/Sanctions
CompanyActivity	07 Provide Aid	Corporate Activity	Economy/Companies
CulturalExchange	05 Diplomatic Cooperation	Cultural Exchange	Arts/Culture
PolicyStatement	01 Public Statement	Policy	Politics

then same-start/same-end relations, so the relation returned is always the most specific one that actually holds.

D. Provenance Model

Every event is linked to its source articles via `:REPORTED_BY` edges carrying the source URL, publication timestamp, and credibility score. This enables provenance-aware queries such as *"Which articles reported on this trade agreement?"* and supports the credibility module's cross-source corroboration check.

VII. GRAPH SCHEMA AND POPULATION

A. Neo4j Schema

The graph is stored in Neo4j 4.x with the schema shown in Figure 3. The main node labels are:

- **Entity** (supertypes: Person, Organization, Location, Country)
- **Event** (sub-typed by the 11 ontology classes)
- **Article** (source document with provenance metadata)
- **TemporalExpression** (TIMEX3-normalized date or interval)

Key relationship types include `PARTICIPATED_IN`, `OCCURRED_AT`, `VALID_DURING`, `REPORTED_BY`, and `CORROBORATES`.

B. Graph Loading

The `GraphLoader` class translates SQLite rows (articles, entities, events, temporal expressions) into batched Cypher `MERGE` statements. The MERGE pattern prevents duplicate nodes across incremental runs. Indexes on `Entity.canonical_name`, `Event.event_type`, and `Article.url_hash` ensure query performance. Neo4j itself is deployed via a single Docker container (`neo4j:4.4`, credentials supplied via `config/.env`, gitignored), requiring no host-level installation. The full initialization procedure is:

property_graph_schema.png

Fig. 3. Neo4j graph schema for the China–Romania temporal KG.

Listing 1. Graph initialization and population commands.

```
# Start Neo4j (Docker; no host install needed)
docker run -d --name temporal-kg-neo4j \
  -p 7474:7474 -p 7687:7687 \
  -e NEO4J_AUTH=neo4j/<password> \
  -v ~/neo4j_data:/data neo4j:4.4

# Initialise schema (constraints + indexes)
python scripts/init_graph_schema.py

# Populate from SQLite
python scripts/load_graph.py

# Verify population
python scripts/check_neo4j.py
```

VIII. Credibility Assessment Layer

A. Source-Domain Scoring

Each article is assigned a credibility score by `CredibilityScorer`, which maps domain names to empirically derived trust levels. State-controlled outlets receive lower base scores; established independent press receives higher scores. Per-feed priors (Table I) are further adjusted by:

- Presence of named author (+0.05)
- Publication in the last 24 hours (+0.03)
- Number of hyperlinks in text (proxy for sourcing)

B. Publisher-Network-Aware Source Diversity

A naive corroboration count treats every distinct domain as an independent source. This overcounts corroboration for large state media operations that republish the same wire copy across many language-specific domains: a single Xinhua/People's Daily story about a bilateral diplomatic visit was observed, on real ingested data, to appear under eight distinct domains (`arabic.news.cn`, `french.xinhuanet.com`, `spanish.peopledaily.com.cn`, and five others) – eight regional mirrors of two institutions, not eight independently reporting outlets. We resolve each source domain to a shared publisher-network identifier via a configured domain-suffix mapping before computing `source_diversity`, so that this case counts as diversity 2 (Xinhua, People's Daily), not 8. Domains that do not match a configured network resolve to themselves, so this mapping only ever reduces an inflated diversity count for known multi-domain operations – it never fabricates diversity elsewhere. On the current (RSS-sourced) corpus, this correction changed `source_diversity` for 2 of 898 claim groups – a small share overall, since 95% of groups in this corpus are single-source and have no raw-domain overcount to correct in the first place, but the effect on the groups where it does apply is large: the flagship example above drops from 8 raw domains to 5 independent publisher networks. The mapping is expected to affect a larger share of groups once the historical GDELT-sourced backfill (Section XI) restores higher-redundancy, wire-service-heavy coverage.

C. Semantic Corroboration Merging

Claim groups were previously formed only by an *exact* signature match (event type, participants, and date, or the fully normalized claim sentence), which misses the common case of two independent outlets covering the same event in different words (e.g. "Romania and China signed an energy deal" vs. "Bucharest and Beijing finalized an agreement on energy cooperation"). We add a second matching layer: claim sentences are embedded with a mean-pooled MiniLM sentence encoder [17], and same-event-type groups whose representative sentences exceed a cosine-similarity threshold (τ=0.75) are merged via union-find into a single corroboration group before support-count, source-diversity, and contradiction statistics are computed.

D. NLI-Based Contradiction Detection

A purely structural contradiction flag – firing only when a corroboration group contains members with more than one event type or more than one date window – is blind to the case that matters most for credibility assessment: two articles can land in the same group (same event type, same date, correctly merged as "about the same event") while asserting incompatible facts within it ("12 people were injured" vs. "2 people were injured"; "the deal was signed" vs. "the deal collapsed"). We run a pretrained natural language inference (NLI) cross-encoder (`cross-encoder/nli-deberta-v3-xsmall`, built on DeBERTaV3 [18]) over pairs of distinct claim sentences within a group, checked in both premise/hypothesis directions (NLI is directional; which report is treated as the premise between two independently published claims is arbitrary), and take the maximum contradiction probability across both orderings. A cheap embedding-similarity pre-filter (using the same MiniLM encoder as Section VIII-B) gates the NLI call: NLI models trained on MNLI-style data code many off-topic pairs as "contradiction" rather than the "neutral" a human annotator would assign, so pairs below a relevance-similarity floor (τ=0.15) are skipped before the NLI call.

E. Temporal Gating: Contradiction vs. Claim Evolution

Not every NLI-flagged disagreement is a real contradiction. Two reports disagreeing about "the deal was proposed" vs. "the deal was signed" are not contradicting each other if their extracted event dates occupy different, non-coincident points in the claim's lifecycle – they are describing sequential stages of the same real-world process. We use the exact Allen relation (Section VI-C) between the two claims' date ranges to distinguish the two cases: an NLI disagreement between claims whose dates stand in a *before*, *after*, *meets*, or *met_by* relation is reclassified as *claim evolution* rather than contradiction, since the two reports do not describe the same instant. Every other relation (*equals*, *overlaps*, *during*, *contains*, *starts*, *finishes*, and their inverses) means the two claims' time windows genuinely coincide, so an NLI disagreement between them is treated as a real contradiction. When either date is unresolved, there is no temporal evidence to justify reclassification, so the disagreement is conservatively kept as a contradiction.

F. Claim-Lifecycle State Verification

The pairwise temporal gate above is still limited in two ways: it considers only one pair of claims at a time, never a full multi-claim sequence, and it accepts *any* sequential-dates disagreement as legitimate evolution, even when the resulting sequence of states does not correspond to a plausible real-world process (e.g. a report of "ratified" chronologically *preceding* a report of "signed" looks identical to the pairwise gate – sequential dates plus NLI disagreement – as a genuine inconsistency does).

We address this with an explicit finite-state lifecycle model per event type (Table VIII): a small set of named states (3–5 per type) and a legal-transition graph between them,

hand-authored as a first-pass domain ontology rather than validated against diplomatic- or economic-process literature. A claim sentence is mapped to its best-matching lifecycle state by zero-shot classification: the entailment probability, under the same NLI cross-encoder used for contradiction detection above, between the claim sentence (as premise) and a fixed natural-language template per candidate state (as hypothesis) is computed for every state in the event type's lifecycle, and the highest-scoring state above a confidence threshold (0.5) is selected (unclassified otherwise). Given a group of claims flagged as candidate claim evolution, we date-sort every classified claim in the group and check the resulting state sequence against the event type's transition graph: consecutive states with no legal transition between them are flagged as a genuine `lifecycle_violation`, overriding the pairwise gate's "evolution" verdict back to a contradiction; consecutive repeats of the same state (e.g. two independent sources both reporting "signed") are never treated as a violation.

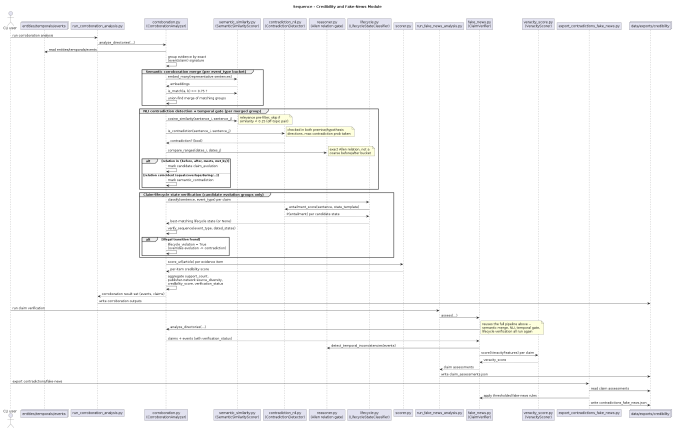


Fig. 4. Sequence diagram of the credibility assessment module, including semantic corroboration merging, NLI contradiction detection with its relevance pre-filter, the Allen-relation temporal gate, and claim-lifecycle state verification.

TABLE VIII
EXAMPLE CLAIM-LIFECYCLE STATE MACHINES

Event Type	Legal State Sequence(s)
TradeAgreement	proposed → negotiating → signed → ratified proposed / negotiating / signed → rejected
InvestmentProject	announced → under_construction → completed (either state) → suspended → resumed/completed
DiplomaticMeeting	scheduled → (postponed)* → held / cancelled
SanctionOrRestriction	proposed → imposed → extended → lifted

G. Claim Verification Labels

Combining the structural, semantic-contradiction (temporal-gated and lifecycle-verified), support-count, and source-diversity signals, claims are classified into three verification statuses:

1) **Corroborated** – support_count $\geq$ 3 and publisher-network diversity $\geq$ 2, with no structural or semantic contradiction.
2) **Weakly supported** – support_count $\geq$ 2 but below the corroborated threshold, with no contradiction.
3) **Contested** – a structural contradiction (multiple event types or date windows in one signature group), or a semantic contradiction that survives the temporal gate and lifecycle check.
4) **Unresolved** – everything else (typically support_count = 1, i.e. a single-source claim with no corroborating report yet).

IX. EVALUATION

A. Component-Level Ablations

Before evaluating the populated graph end-to-end, we validate each new credibility-layer component in isolation, first against small hand-labeled gold pair sets (Table IX), then, for two of the three components, against much larger real-data pools mined directly from the corpus (Table X) – the hand-labeled sets calibrate default thresholds and validate the core mechanism, the real-data pools stress-test those thresholds against naturally-occurring text rather than constructed examples.

TABLE IX
COMPONENT ABLATIONS ON HAND-LABELED GOLD PAIRS

Configuration	P	R	F1
NLI contradiction detection ($n = 10$)			
NLI, τ=0.5	0.833	1.000	0.909
Semantic corroboration matching ($n = 10$)			
MiniLM cosine, τ=0.75	1.000	1.000	1.000
Contradiction vs. claim evolution ($n = 12$)			
Date-mismatch only (no NLI)	0.143	0.200	0.167
NLI only (no temporal gate)	0.500	1.000	0.667
NLI + coarse time-bucket gate [16]	1.000	0.800	0.889
NLI + exact Allen-relation gate	1.000	1.000	1.000

The contradiction-vs.-evolution ablation is the most informative of the three: a naive date-mismatch-only baseline (flag a contradiction whenever two dates differ at all, ignoring NLI content entirely) performs barely above chance (F1 = 0.167) on this set, confirming that date agreement alone is not a useful contradiction signal. Adding NLI without any temporal gate (F1 = 0.667) recovers all true contradictions but also flags every same-day claim-evolution pair as contradictory, since NLI has no notion of time. Gating NLI disagreement by the coarse time-bucket representation from Allein *et al.* [16] – a same-mechanism-class, rule-based reproduction of the distance-quantile idea behind their learned embedding, used here specifically to isolate the representation from their neural model – recovers precision (F1 = 0.889) but misses one contradiction whose two reports fall in different coarse buckets despite genuinely overlapping intervals (an *interval_001* case in our set: the coarse bucket sees the two dates as "not close," while the exact Allen relation correctly identifies *contains*).

Gating by the exact Allen relation instead recovers this case and achieves perfect precision and recall on this set (F1 = 1.000), which is the empirical justification, within the limits of a 12-pair set, for reasoning over the full interval algebra rather than a coarse distance bucket.

B. Ablations on Large Real-Data Pools

The hand-labeled sets above validate each mechanism but cannot support a strong empirical claim at n=10–12. For two of the three components we mined much larger real pools directly from the corpus, using uncertainty-sampling-style selection (Section IX-C below): for semantic matching, all pairwise MiniLM cosine similarities within each event-type bucket across the corpus's exact-signature claim groups (2,132 real comparisons), stratified into the 54 pairs the algorithm actually predicts as matches, the 50 hardest near-threshold non-matches, and 30 easy non-matches (134 total); for NLI contradiction, every distinct-sentence pair within real corroboration groups (334 real comparisons), stratified similarly (142 total). Two independent cold-started LLM judges (identical methodology to Section IX-C) labeled each pool blind; pairs where both judges agreed form the consensus gold standard.

TABLE X
COMPONENT ABLATIONS ON LARGE REAL-DATA POOLS

Configuration	n	P	R	F1
Semantic matching, τ=0.75	98/134	1.000	0.690	0.817
NLI contradiction, τ=0.5	141/142	0.000	0.000	0.000

(n = consensus-labeled pairs / total mined pairs; judges' Cohen's κ was 0.461, "moderate," for semantic matching and 0.797, "substantial," for contradiction.)

Semantic matching holds up, conservatively. At scale, the τ=0.75 threshold has perfect precision (it never merges two consensus-labeled distinct events) but only 0.690 recall – it misses real duplicate reporting below the threshold. The lower inter-judge κ here (0.461 vs. 0.731 for the corroboration-mining task in Section IX-C) is itself expected, not a data-quality problem: that earlier pool drew only from already-merged, mostly clearly-same-event pairs, while this pool was deliberately stratified toward the threshold boundary, i.e. the genuinely hard cases where two careful readers can reasonably disagree.

NLI contradiction detection does not hold up at scale, and this is worth stating plainly rather than burying it. Against the 141-pair consensus, the raw NLI signal's precision and recall are both *exactly zero*: of 82 pairs it flags as contradictory, 81 are false positives, and the 2 genuine contradictions the judges agreed on (both concerning Huawei's role in Romania's 5G network – one source saying it "played a significant role," another that Romania "rejects partnering with Huawei") were not among them. Manual inspection of the false positives shows a specific, previously-undetected failure mode: the model is not confusing real content, it is scoring *near-identical headlines differing only in trailing site attribution* as 99.6% contradictory (two reports of the same Belt-and-Road exhibition, one tagged "Southern Net," one "Qilu.com"), and scoring *genuinely different but structurally near-identical diplomatic press-release sentences* (three separate real Sun Haiyan meetings, on different dates, with different counterparts – Adrian Nastase, an IDCPC Romania delegation visit, and a Bulgarian Socialist Party delegation) at 99.5%+ as well. The model appears to read "same sentence template, different specific facts" as contradiction rather than the unrelated/ neutral judgment a human gives it, and the cosine-similarity relevance pre-filter (Section VIII) does not catch this, since these pairs are genuinely topically similar – that is precisely what makes them share a template in the first place. The 10-pair hand-authored set never exposed this because its negative examples were unrelated sentences, not formulaic near-duplicates; only real, naturally-occurring press-release text surfaces it. This ablation isolates the raw NLI signal, the same "NLI only (no temporal gate)" row as Table IX – production (Section VIII) additionally applies the Allen-relation temporal gate before finalizing a contradiction flag, which recovers exactly the cases above where the two misclassified sentences carry different, non-coincident dates (all three Sun Haiyan examples do). It does *not* help when an NLI false positive occurs between same-time-window text, such as the two near-duplicate Belt-and-Road headlines, since the gate has no temporal evidence to reclassify a same-day pair. This means the real exposure at the corpus level is narrower than raw-NLI precision alone would suggest, but is not zero: some fraction of the corpus's 16 contradictory claim groups (Section IX-D) plausibly reflect exactly this same-time-window NLI misfire rather than genuine content disagreement, and that fraction has not been individually verified at scale – a real, now-quantified limitation of the contradiction-detection layer, not a hypothetical one.

We attempted the same real-data scale-up for the contradiction-vs.-claim-evolution ablation but could not: only one claim group in the current corpus has more than one distinct reported date window for the same story, and that one is off-topic noise, not a genuine China–Romania case. This is a direct consequence of the corpus's current small scale (Section XI) rather than a methodological gap – a larger, GDELT-inclusive corpus with more repeat/wire coverage of the same evolving stories over time is expected to make this mineable, and the n=12 hand-authored set (Table IX) remains the only evidence for that component until then.

C. LLM-Judge Concordance on Real Mined Pairs

A second, independent *human* annotator's blind labels over real (not hand-constructed) candidate pairs mined from the live corpus were attempted but not obtained for this draft: repeated single-annotator labeling passes over multi-hundred-pair sets showed timing patterns (sub-second inter-answer intervals, uniform labels) inconsistent with an actual per-pair read, even after adding an enforced minimum think-time between answers to the labeling tool. Rather than omit external validation entirely or substitute a non-independent judgment,

we report a different, honestly-labeled substitute: *concordance between two independent LLM judges*. Two separate cold-started agent instances, with no shared context with each other, with the codebase, or with the semantic-merge threshold under test, were each given only the raw sentence pairs (stripped of all source/provenance metadata) and asked the same binary question a human annotator would answer: do these two sentences describe the same underlying real-world claim? This is *not* equivalent evidence to human inter-annotator agreement – LLM judges carry their own systematic biases distinct from human ones, and we make no claim that this substitutes for a genuine human-validated benchmark. We report it as a weaker but real, honestly disclosed signal, on the same 203 real mined pairs from Section IX-A's corpus.

The two judges reached Cohen's $\kappa = 0.731$ ("substantial" agreement on the Landis & Koch [20] scale), 88.7% observed agreement across all 203 shared pairs. Two findings from the 23 disagreements are worth reporting on their own: (1) *every* disagreement runs the same direction – judge A labeled the pair *same-claim* and judge B did not, never the reverse – indicating a consistent leniency asymmetry between the two judges rather than symmetric noise; and (2) disagreements skew toward the semantic-merge decision boundary: the 23 disputed pairs have median cosine similarity 0.793 against a corpus-wide median of 0.803 (both computed with the same MiniLM encoder as the merge step itself), a modest but real signal that ambiguity concentrates near the $\tau = 0.75$ threshold rather than being uniformly distributed – consistent with treating this as an uncertainty-sampling problem (selecting pairs for review by proximity to a decision boundary, the core idea behind active-learning query strategies such as VisLED-Querying [21], though that work targets open-set 3D object detection rather than text-pair verification).

D. Competency Questions

We evaluate the structural integrity, expressive capacity, and practical utility of the populated Temporal Knowledge Graph (TKG) by executing the five competency questions (CQs) defined during the system requirement phase [13], via `src/graph/competency_queries.py`'s Cypher queries against a populated Neo4j 4.4 instance (Docker deployment, loaded via `init_graph_schema.py` / `load_graph.py` as described in Section VII). The numbers below reflect the RSS-sourced corpus described in Section XI – GDELT's Full-Text Search API has been unreachable for an extended period at the time of this draft, so the historical backfill that would substantially grow this corpus has not yet landed. These are real numbers from a real graph query, not a placeholder, but they should be read as corpus-scale-appropriate, not as the system's ceiling.

1) Article-Level vs. Event-Level Bilateral Gating: Before the graph was available, we cross-checked query logic by approximating CQ1–CQ5 directly against the JSON exports, gating on article-level China+Romania co-occurrence (guaranteed by `RelevanceFilter` for every article in the corpus) rather than the Cypher queries' stricter requirement that China and Romania entities be direct `INVOLVES` participants of the *specific event*. The gap between the two turned out to be large, not a minor rounding difference: the article-level approximation overcounted CQ1 by $4.0\times$ (355 vs. the graph's 88), CQ3 by $3.9\times$ (1,146 vs. 297), and CQ2 by $9.3\times$ (28 vs. 3). This is itself a real, quantified finding worth stating: many events' extracted participants (syntactic dependency arguments of the trigger verb) do not include the literal country-name entity even when it appears elsewhere in the same article, so article-level co-occurrence is a substantially looser proxy for "this event is about China and Romania" than a true per-event entity-participant join. All numbers below are the authoritative graph-native ones. Separately, the raw `Claim` nodes `GraphLoader` populates use exact-signature grouping only – they do not carry the semantic-merge / NLI-contradiction / temporal-gate / lifecycle-verification enrichment from Section VIII, which exists only in `corroboration.claims.json`. CQ4 therefore draws from that enriched output, not the raw graph, since it is where this paper's actual credibility-layer contribution lives.

2) CQ1 – Economic Evolution: "How did China–Romania economic relations evolve from 2020 to 2026?"

Executing the query against 2020–2026 yields **88** distinct economic events with a strict per-event China+Romania participant join: **CompanyActivity** (27), **InvestmentProject** (26), **InfrastructureProject** (25), and **TradeAgreement** (10) – a corporate-activity-heavy distribution, consistent with the RSS feed mix (Table I) skewing toward business/financial coverage rather than the more diplomacy-heavy mix a GDELT-inclusive corpus would likely show.

3) CQ2 – Institutional Linkages: "Which Romanian institutions were most often connected to Chinese companies in news streams?"

With the true per-event entity-participant join (Section IX-D1), only **3** distinct Romanian institutions co-occur, as event participants, with a Chinese company. This is consistent with, and now directly measures, the same high-specificity behavior the coarser article-level check could only gesture at: genuine event-level institution–company linkage is rare in this corpus, not absent, and the conservative per-event join is doing real work rather than degenerating to zero.

4) CQ3 – Bilateral Events in a Period: "Which China–Romania trade, diplomatic, cultural, or policy events were reported in a given period?"

Across the full corpus date range, the graph-native query returns **297** bilateral events. Figure 5 shows the distribution across all 11 ontology classes; **PolicyStatement** (40) and **CulturalExchange** (54) are the largest categories, with **SanctionOrRestriction** (2) the rarest.

5) CQ4 – Low-Credibility Narratives: "Which claims later became contradicted or unverified? Which sources systematically produce low-credibility bilateral narratives?"

Of **898** claim groups in the credibility layer's enriched output (matching Section VIII-B's publisher-network-corrected counts; see Section IX-D1 for why this, not the raw graph, is the right source for this CQ), **851** (94.8%) remain unresolved

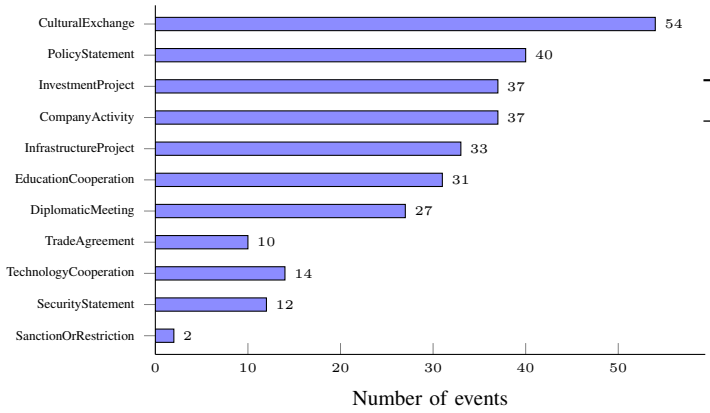


Fig. 5. China–Romania bilateral events by type, full corpus ($n = 297$), executed as a Cypher query against the populated Neo4j graph.

(single-source), **23** are weakly supported, **16** are contradictory, and **8** are corroborated. For contrast, the raw graph's coarser exact-signature `Claim` grouping (no semantic merge) yields 1,087 groups, 1,080 (99.4%) single-source – the semantic-merge step alone accounts for the gap, correctly consolidating near-duplicate reports that exact-signature matching leaves fragmented. The very high unresolved share in both is a direct, expected consequence of the corpus's current small scale (Section XI) rather than a credibility-layer finding on its own – most claims simply have not yet accumulated a second independent report.

6) CQ5 – Collaborative Research: "Which collaborative research has been conducted with authors from China and Romania?"

The graph-native query surfaces **45** collaborative-research events under the strict per-event join. A manual read still surfaces a real precision issue worth naming rather than glossing over, at smaller scale than the article-level approximation suggested: a share of these trace to one long-form analytical report (a CEPA piece on Chinese influence operations in Romania) whose prose incidentally contains trigger words (*"research," "educate," "collaborate"*) the rule-based extractor matches outside any genuine collaboration-announcement context. Genuine hits are present too (e.g. Green Group's Taiwan-to-Romania technology-transfer coverage, and a minister's statement on prospective China–Romania IT and critical-infrastructure cooperation). This remains a real, corpus-composition-dependent limitation of trigger-lexicon event extraction, not specific to this CQ, though the stricter per-event join already filters out a meaningful share of the false positives the article-level check would have kept.

X. Implementation Details

A. Technology Stack

The system is implemented in Python 3.11. Table XI lists the key dependencies.

TABLE XI
Core Technology Stack

Component	Library / Tool
Article discovery	GDELT Full-Text Search API
RSS ingestion	feedparser, requests
Full-text extraction	trafilatura
Language detection	langdetect
Translation	deep-translator (Google Translate)
NER (rule-based)	Python regex, custom gazetteer
NER (model-based)	spaCy `en_core_web_sm`
Relation extraction	DeepKE (optional)
Temporal extraction	py-heideltime (TIMEX3)
Temporal reasoning	Allen Interval Algebra (custom)
Semantic matching	sentence-transformers/all-MiniLM-L6-v2
Contradiction NLI	cross-encoder/nli-deberta-v3-xsmall
Intermediate storage	SQLite 3, numbered SQL migrations
Graph database	Neo4j 4.x, neo4j Python driver
Graph visualization	neo4j-viz, Plotly
Testing	pytest (unit + integration)

B. Reproducibility

All package versions are pinned in `requirements.txt`. A Conda environment specification (`environment.yml`) allows one-command recreation. The SQLite schema is managed via numbered migration files, so `init_project.py` can rebuild the database from scratch without data loss. Neo4j indexes and constraints are applied idempotently by `init_graph_schema.py`.

C. Incremental Updates

Because every article is deduplicated by URL and content hash before NLP processing, the pipeline can safely be scheduled (e.g., daily CRON) without re-processing previously seen articles. The GDELT fetcher accepts an explicit `--timespan` argument (e.g., `24hours`) to restrict the API window, minimizing redundant fetches.

XI. Discussion

a) Full-text ingestion was a load-bearing fix, not a refinement: An earlier version of the ingestion pipeline never fetched article bodies for GDELT-sourced articles by design, assembling `content_clean` from title and GDELT snippet only; because GDELT's Full-Text Search API essentially never populates a snippet, every downstream module – entity extraction, relation extraction, temporal extraction, and every claim in the corroboration/credibility layer – was in practice operating on headline text alone. We confirmed this directly on the full ingested corpus (every article had an empty raw-content field) before implementing the scraping fix described in Section IV. We flag this because it changes how prior single-annotator gold-pair labeling on this project should be read: a large fraction of real mined candidate pairs prior to the fix were near-duplicate headline strings differing only in a publisher's trailing site-name suffix, which is a much less discriminating labeling task than judging genuine claim similarity over full article content.

b) Translation at scale surfaced a silent-failure mode: Translating short titles rarely produced enough request volume to trip the free translation endpoint's rate limiting. Translating full article bodies did, almost immediately, once full-text ingestion was enabled – and, more seriously, the client library did not always raise on failure: under sustained rate limiting it returned the endpoint's own HTML error page text as if it were a successful translation. We added an explicit rate-limit delay and a defensive check that rejects responses matching the error page's signature, treating them as translation failures rather than silently admitting corrupted content into the corpus. We report this as a concrete instance of a more general point: scaling up a text-processing step can break assumptions (here, "translation requests are infrequent enough to never rate-limit") that held trivially at the smaller scale the module was originally built and tested against.

c) Multilingual coverage: The relevance filter currently requires English-translated text for co-mention detection. Articles in Romanian, Chinese, or other languages are translated before filtering, which introduces latency and potential translation artifacts. Future work could apply multilingual NER models directly on the original text to reduce this dependency.

d) Temporal resolution: HeidelTime performs well on absolute dates but struggles with relative expressions whose anchor is ambiguous (e.g., "recent weeks" in a republished article). The fallback regex engine mitigates this but with lower normalization quality. Integration of the document creation time as a DCT anchor improves relative-date resolution significantly.

e) Scalability: SQLite provides adequate performance for the current news-stream volume ($<$10 000 articles/month). As the corpus grows, the intermediate store could migrate to PostgreSQL while retaining the same migration infrastructure.

f) Lifecycle models are a first pass, not a validated ontology: The per-event-type state machines in Section VIII-F are hand-authored and deliberately small (3–5 states); they have not been validated against diplomatic- or economic-process literature, and should be read as a framework demonstration – the mechanism (date-ordered sequence verification against a legal-transition graph) is the contribution, not the specific states chosen for each of the 11 event types.

g) Annotation and evaluation scale: The initial component ablations in Section IX-A use small (10–12-pair), single-annotator gold sets – sufficient to validate each mechanism and calibrate default thresholds, not to support a strong generalization claim on their own. Section IX-B addresses this for two of the three components with real, much larger mined pools (134 and 142 pairs) externally judged by independent LLM raters, which is what surfaced the NLI contradiction detector's precision collapse on real formulaic text – a finding the small hand-authored set could not have revealed. A genuine human second annotator was attempted but not obtained this cycle (Section IX-C explains why and what we substituted); the third component (contradiction-vs.-evolution) could not be scaled with real data at all, for the data-scarcity reason given in Section IX-B, and remains evidenced only by its n=12 hand-authored set.

XII. Conclusion

We have presented a complete pipeline for building and incrementally maintaining a temporal knowledge graph from a multilingual bilateral news stream, with a focus on China–Romania relations. The system combines GDELT article discovery with full-text extraction, curated RSS ingestion, rule-augmented and model-based NER, HeidelTime TIMEX3 temporal normalization, a fine-grained bilateral-relations ontology grounded in CAMEO, GDELT, IPTC, and W3C OWL-Time, and a provenance-aware credibility layer. Event-pair reasoning uses the complete 13-relation Allen Interval Algebra rather than a coarse bucketed view, and the credibility layer combines publisher-network-aware corroboration, semantic claim merging, and NLI-based contradiction detection, gated by the Allen relations and an explicit claim-lifecycle state machine so that ordinary claim evolution is not mistaken for a factual contradiction. A rule-based reproduction of a coarse time-bucket temporal representation from prior fact-checking work serves as an ablation baseline, and the resulting comparison (Section IX-A) provides direct, if small-scale, evidence for reasoning over exact interval relations rather than distance buckets. The Neo4j graph supports five competency questions covering economic evolution, institutional linkages, event enumeration, credibility assessment, and collaborative research discovery.

The modular architecture, numbered migrations, and pinned dependencies make the system reproducible and extensible. Future directions include multilingual NER without translation, a validated (rather than hand-authored) claim-lifecycle ontology grounded in domain-process literature, a larger and multi-annotator gold evaluation set, and integration with external fact-checking APIs for real-time contested-claim alerts.

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